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SECTION 1: DDL

Q1 – CREATE TABLE

```
MySQL 8.0 Command Line Client
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 21
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bank;
Query OK, 1 row affected (0.01 sec)

mysql> USE bankdb;
Database changed
mysql> CREATE TABLE bank_transactions (
  ->     txn_id INT PRIMARY KEY,
  ->     customer_name VARCHAR(50),
  ->     branch_name VARCHAR(50),
  ->     transaction_type VARCHAR(20),
  ->     amount DECIMAL(10,2),
  ->     transaction_date DATE
  -> );
ERROR 1050 (42S01): Table 'bank_transactions' already exists
mysql> SHOW TABLES;
+-----+
| Tables_in_bankdb |
+-----+
| bank_transactions |
+-----+
1 row in set (0.04 sec)

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id     | int       | NO   | PRI | NULL    |       |
| customer_name | varchar(50) | YES  |     | NULL    |       |
| branch_name | varchar(50) | YES  |     | NULL    |       |
| transaction_type | varchar(20) | YES  |     | NULL    |       |
| amount     | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date      | YES  |     | NULL    |       |
| account_no | varchar(20) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.01 sec)
```

Q2 – ALTER TABLE ADD COLUMN

```
mysql> ALTER TABLE bank_transactions
-> ADD account_no VARCHAR(20);
ERROR 1060 (42S21): Duplicate column name 'account_no'
mysql> ALTER TABLE bank_transactions
-> ADD account_no VARCHAR(20);
ERROR 1060 (42S21): Duplicate column name 'account_no'
mysql> DESCRIBE bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(50)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

```
7 rows in set (0.00 sec)
```

Q3 – ALTER TABLE MODIFY COLUMN

```
mysql> ALTER TABLE bank_transactions
-> MODIFY customer_name VARCHAR(100);
Query OK, 10 rows affected (0.17 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql>
```

Q4 – RENAME TABLE

```
mysql> RENAME TABLE bank_transactions TO customer_transactions;
Query OK, 0 rows affected (0.03 sec)

mysql>
```

Q5 – TRUNCATE TABLE

```
mysql> TRUNCATE TABLE customer_transactions;
Query OK, 0 rows affected (0.06 sec)

mysql>
```

Q6 – DROP TABLE

```
mysql> CREATE TABLE bank_backup AS  
-> SELECT * FROM customer_transactions;  
Query OK, 0 rows affected (0.04 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

Conclusion of section 1

```
mysql> SHOW TABLES;
```

```
+-----+  
| Tables_in_bankdb |  
+-----+  
| bank_backup      |  
| customer_transactions |  
+-----+  
2 rows in set (0.00 sec)
```

```
mysql> DESCRIBE customer_transactions;
```

```
+-----+-----+-----+-----+-----+-----+  
| Field          | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| txn_id         | int           | NO   | PRI | NULL    |       |  
| customer_name  | varchar(100)  | YES  |     | NULL    |       |  
| branch_name    | varchar(50)   | YES  |     | NULL    |       |  
| transaction_type | varchar(20)   | YES  |     | NULL    |       |  
| amount         | decimal(10,2) | YES  |     | NULL    |       |  
| transaction_date | date          | YES  |     | NULL    |       |  
| account_no     | varchar(20)   | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
7 rows in set (0.02 sec)
```

SECTION - 2

Q7 – INSERT

```
mysql> INSERT INTO customer_transactions
-> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
-> VALUES
-> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
-> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
-> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
-> (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
-> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
-> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
-> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
-> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
-> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
-> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
Query OK, 10 rows affected (0.03 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> _
```

Q8 – UPDATE

```
mysql> UPDATE customer_transactions
-> SET amount = 5000
-> WHERE txn_id = 105;
Query OK, 1 row affected (0.09 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> _
```

```
mysql> RENAME TABLE customer_transactions TO bank_transactions;
Query OK, 0 rows affected (0.05 sec)

mysql> _
```

Q9 – DELETE

```
mysql> ^C
mysql> DELETE FROM bank_transactions
    -> WHERE txn_id = 111;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

Q10 – SELECT

```
mysql> SELECT *
    -> FROM bank_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

```
10 rows in set (0.00 sec)

mysql>
```

Q11 – SELECT WITH WHERE

```
mysql> SELECT *
    -> FROM bank_transactions
    -> WHERE transaction_type='Deposit';
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

```
6 rows in set (0.00 sec)

mysql>
```

Q12 – SELECT WITH ORDER BY

```
mysql> SELECT *
-> FROM bank_transactions
-> ORDER BY amount DESC;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
```

SECTION 3

Q13 – GRANT SELECT

```
mysql> GRANT SELECT
-> ON bank_transactions
-> TO Auditor1;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> GRANT SELECT
-> ON bank_transactions
-> TO Auditor1;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.06 sec)

mysql> GRANT SELECT
-> ON bank_transactions
-> TO Auditor1;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> GRANT SELECT
-> ON bankdb.bank_transactions
-> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

Q14 – GRANT ALL PRIVILEGES

```
mysql> GRANT ALL
-> ON bank_transactions
-> TO BranchManager;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT ALL PRIVILEGES
-> ON bankdb.bank_transactions
-> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql>
```

Q15 – REVOKE SELECT

```
mysql> REVOKE SELECT
  -> ON bank_transactions
  -> FROM Auditor1;
ERROR 1147 (42000): There is no such grant defined for user 'Auditor1' on host '%' on table 'bank_transactions'
mysql> REVOKE SELECT
  -> ON bankdb.bank_transactions
  -> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql>
```

Q16 – REVOKE ALL

```
mysql> REVOKE ALL
  -> ON bank_transactions
  -> FROM BranchManager
  -> REVOKE ALL
  -> ON bank_transactions
  -> FROM BranchManager;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server
version for the right syntax to use near 'REVOKE ALL
ON bank_transactions
FROM BranchManager' at line 4
mysql> REVOKE ALL PRIVILEGES
  -> ON bankdb.bank_transactions
  -> FROM 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.01 sec)
```

SECTION 4

Q17 – COMMIT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
  -> SET amount = 6000
  -> WHERE txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
```

Q18 – SAVEPOINT

```
mysql>
mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

Q19 – ROLLBACK TO SAVEPOINT

```
mysql> UPDATE bank_transactions
      -> SET amount = 99999
      -> WHERE txn_id = 102;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> SELECT * FROM bank_transactions
      -> WHERE txn_id = 102;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
102	Sita	Hyderabad	Withdrawal	99999.00	2024-01-06	NULL

1 row in set (0.01 sec)

Q20 – FULL TCL EXAMPLE

```
mysql>
mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
      -> SET amount = 9000
      -> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
```